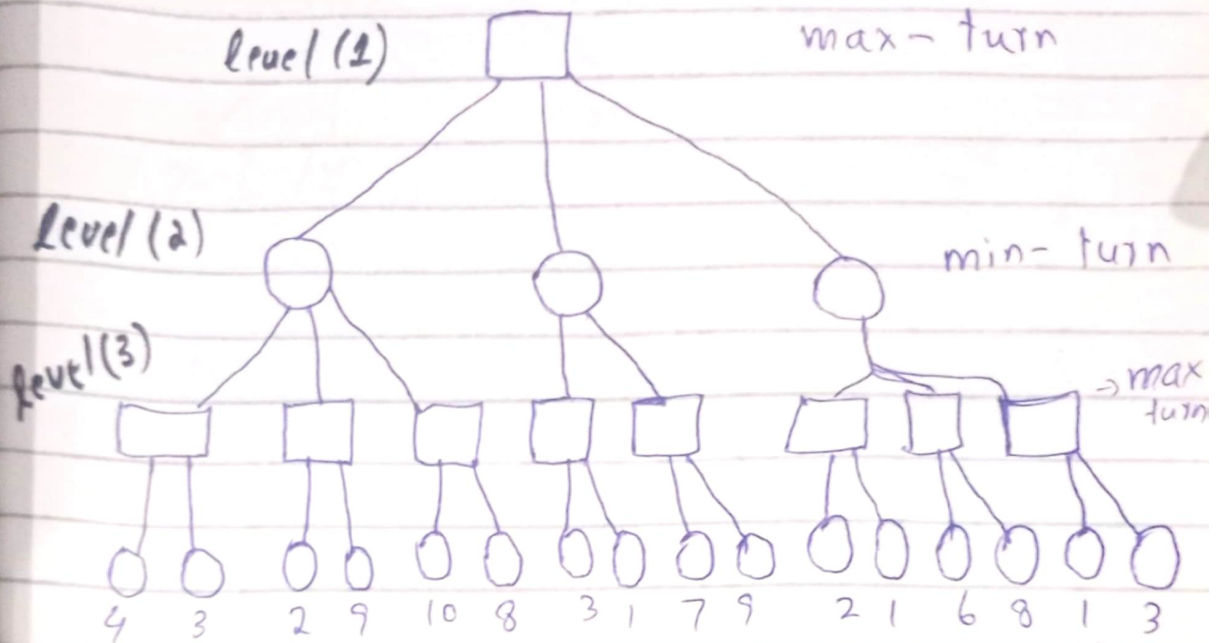
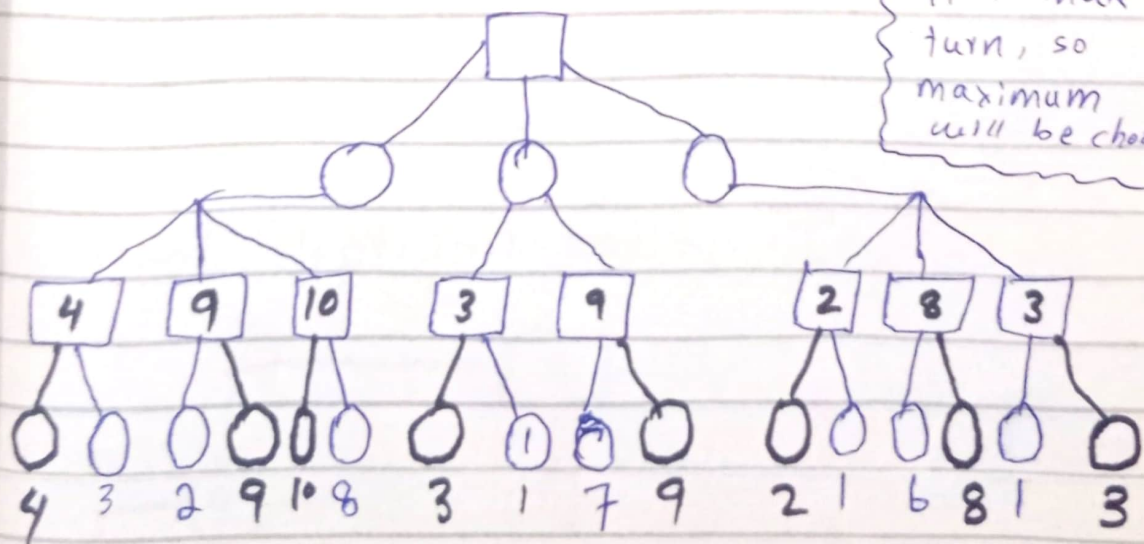


# (QUESTION 1 MINMAX)

Initial Tree:

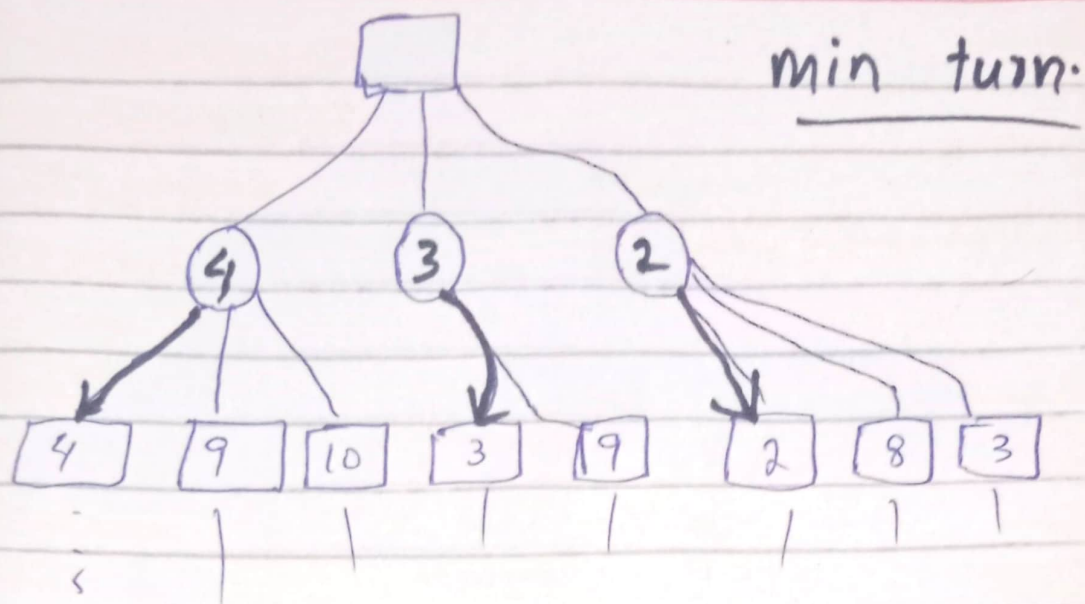


Step: 1 starting from leaf nodes  
level 3



Step (2) Level: 2 It is min turn.

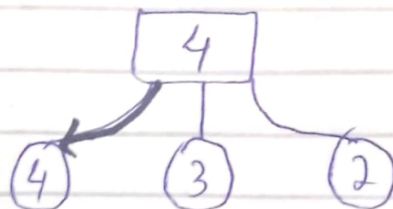
- min has 8 options, 3 moves.
- min will choose least (minimum)
- so it chooses 4, 3, 2.



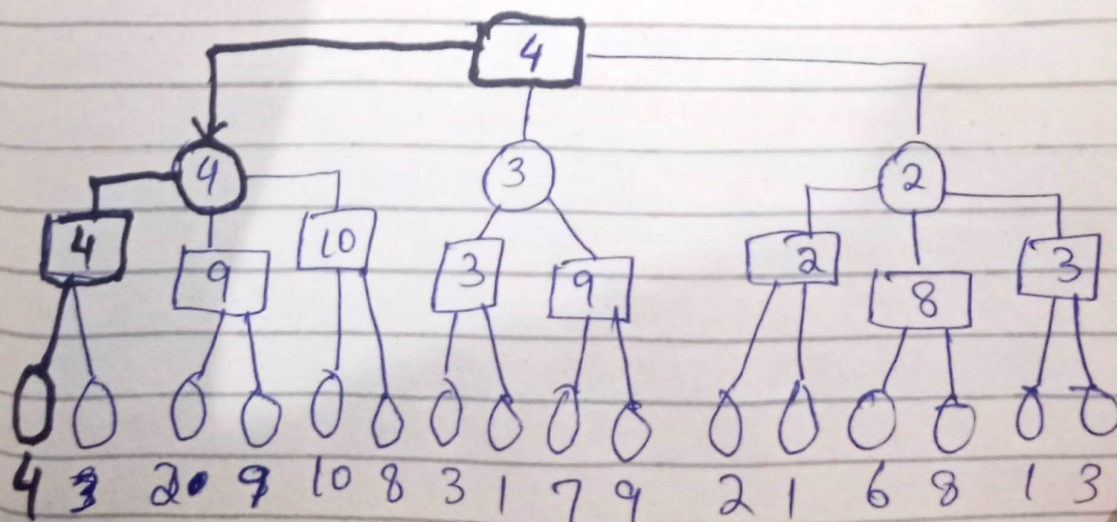
Step (3) Max-turn: level (1)

⇒ max has 1 option out of 3 possible moves.

max will choose  $\max(4, 3, 2)$   
so 4 it is.

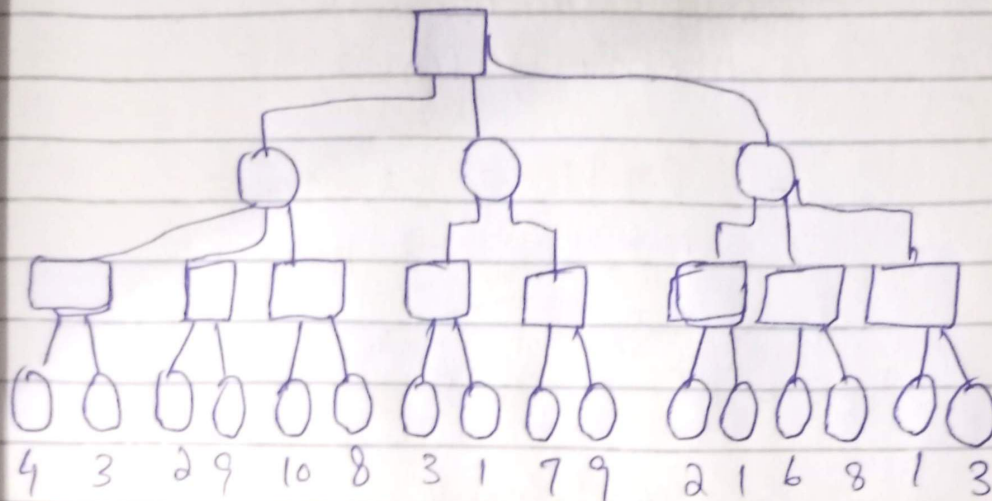


final dept-first-search tree.



## QUESTION : 2 Alpha Beta:

initial tree:

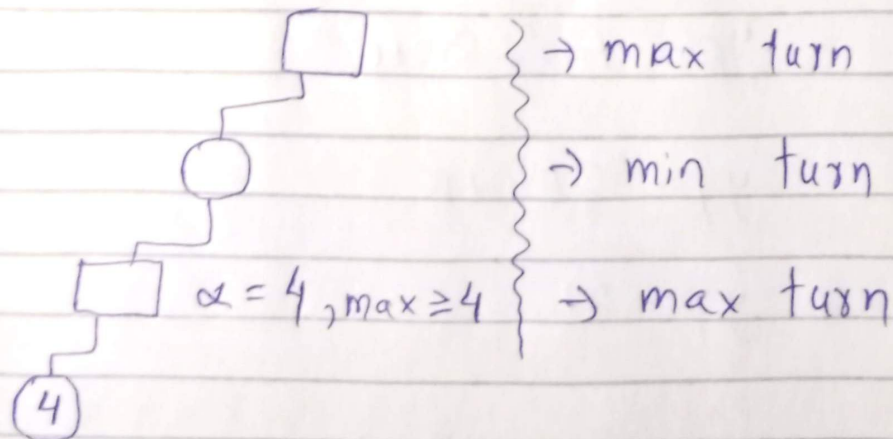


Step = 1:

DFS

L.H.S:

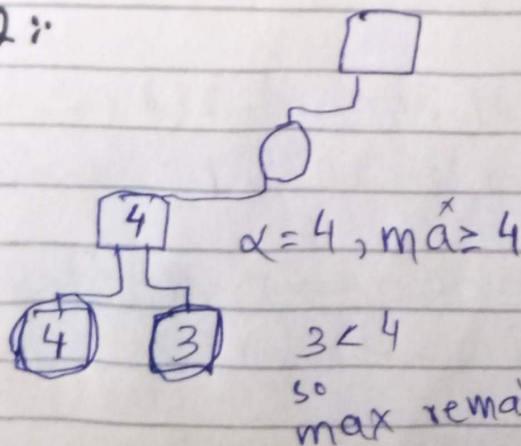
max, turn



Step : 2:

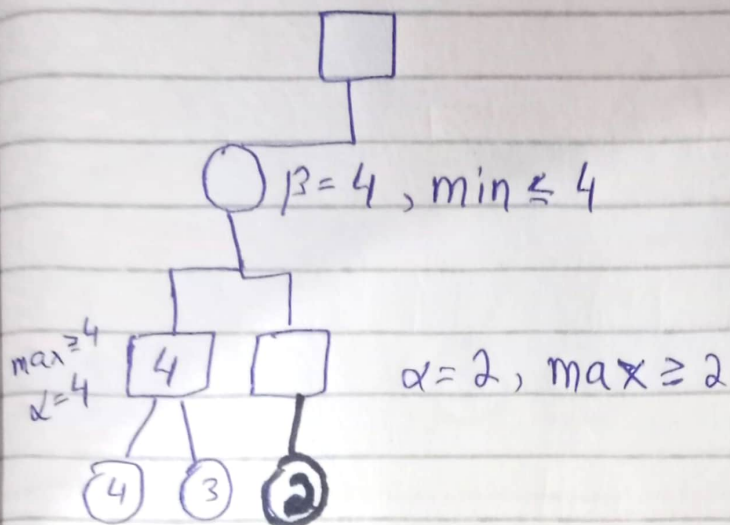
{ max  
turn }

exploring  
2nd leaf:

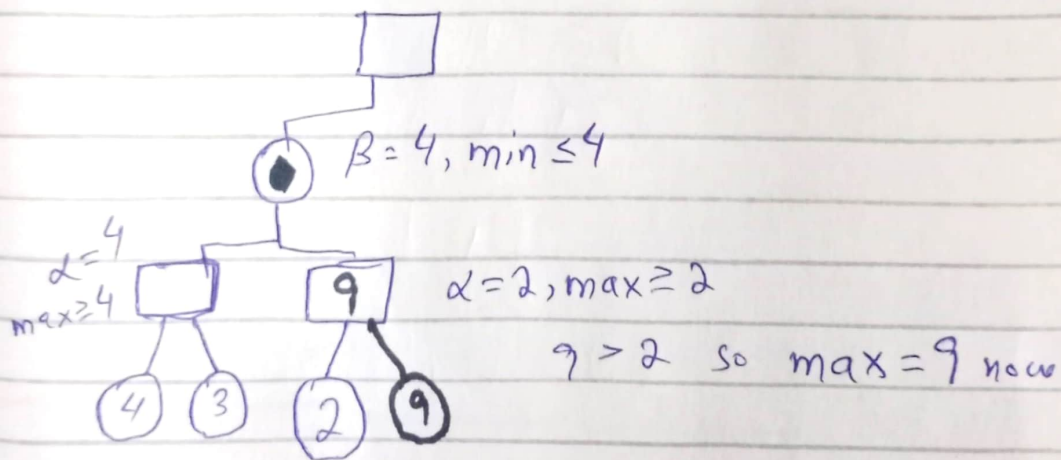


Step 3:

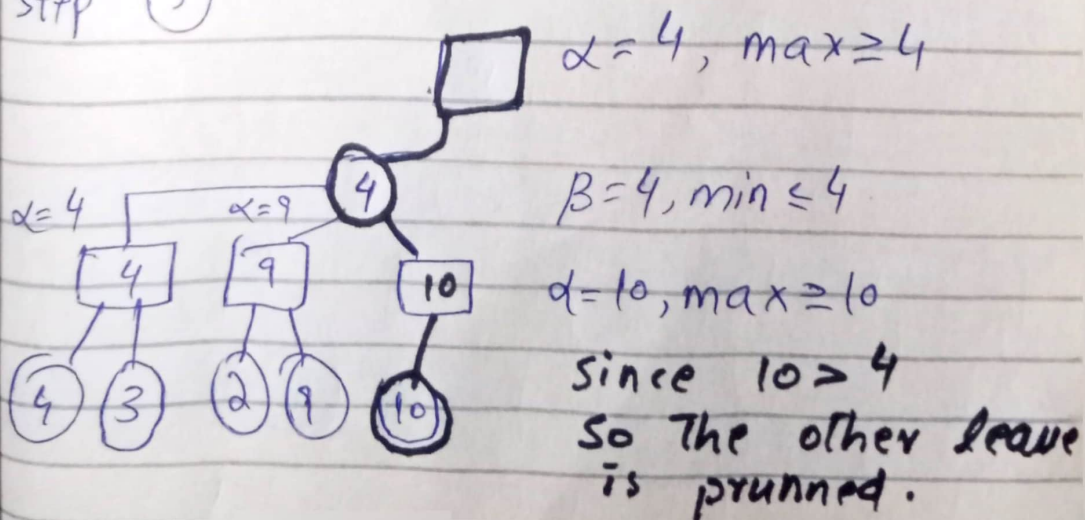
~~Pruned~~



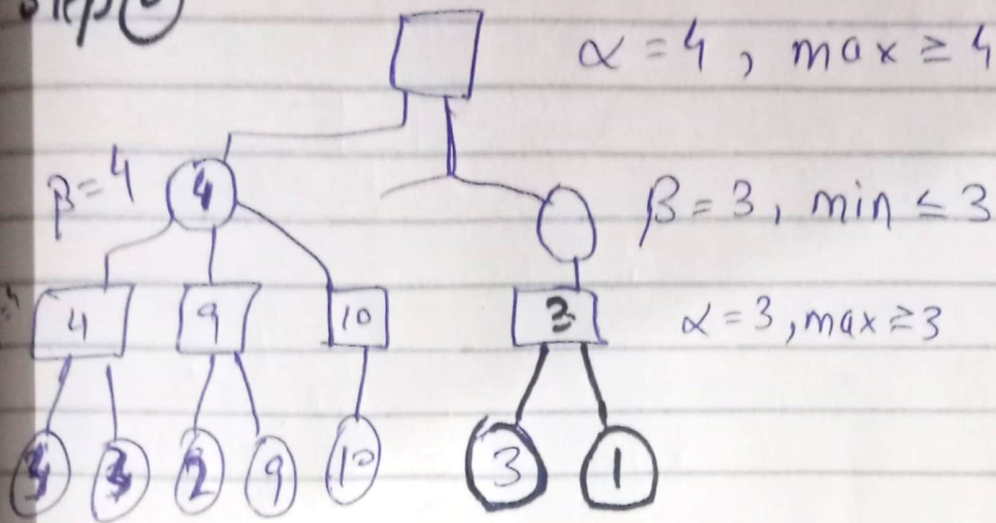
Step 4:



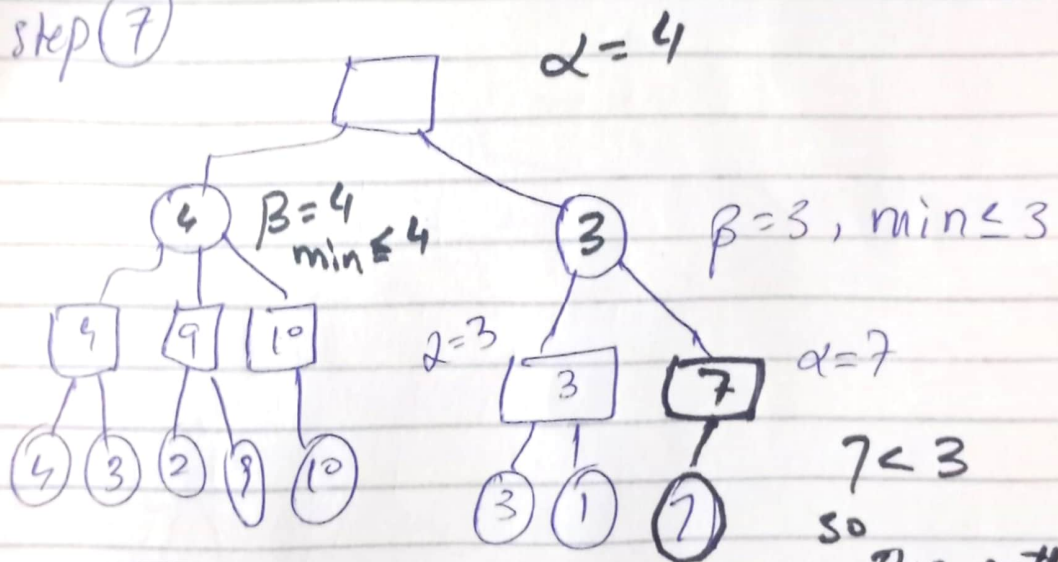
Step 5



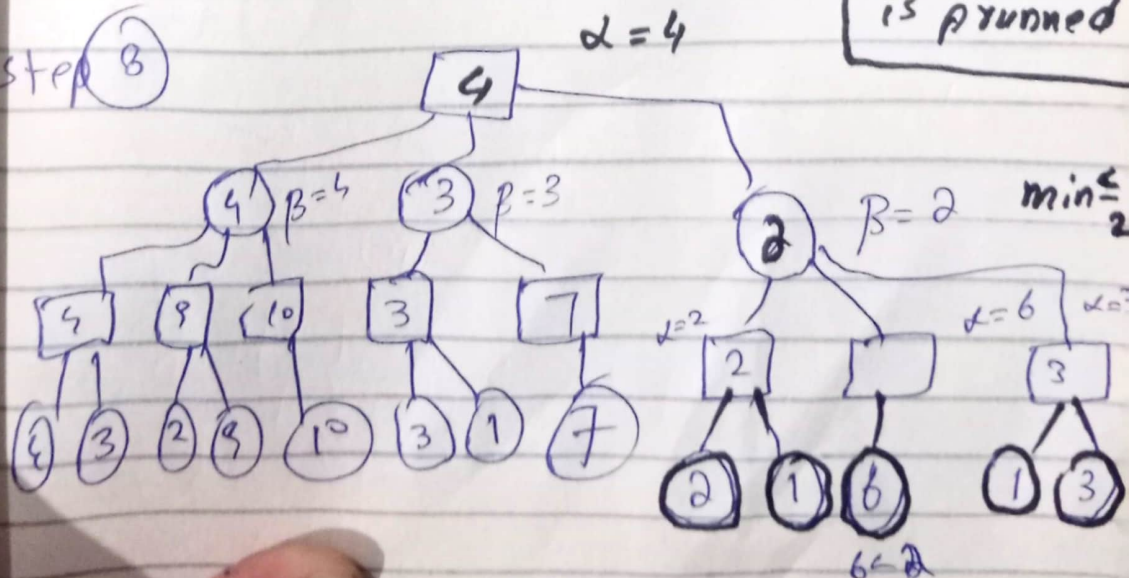
Step (6)



Step (7)



Step (8)



~~when~~ Last step:

So the final value of  
root node = 4.

and dfs tree is the  
left most branch of the  
tree.

